TABLA DE DERIVADAS E INTEGRALES

DERIVADAS

1) $y = cte$	$y' = 0$
2) $y = x$	$y' = 1$
3) $y = c.x$	$y' = c$

Derivada de una función

$$\lim_{x \to a} f(x) = f'(x) = \frac{dy}{dx}; a = \mathbb{R}; y = f(x)$$

Derivada de una función de función

$$y = f(u); u = f(x)$$

2) $y = u$	$y' = u'$
3) $y = c.u$	$y' = c.u'$
4) $y = u \pm v \pm w$	$y' = u' \pm v' \pm w'$
5) $y = u.v$	$y' = u'.v + u.v'$
6) $y = \frac{u}{v}$	$y' = \frac{u'.v - u.v'}{v^2}$
7) $y = u^v$	$y' = u^v \left[v' \ln(u) + v.\frac{u'}{u} \right]$

8) $y = x^m$	$y' = m.x^{m-1}$
9) $y = \sqrt[m]{x}$	$y' = \frac{1}{m\sqrt[m]{x^{m-1}}}$
10) $y = \sqrt{x}$	$y' = \frac{1}{2\sqrt{x}}$
11) $y = \log_a(x)$	$y' = \frac{1}{x}\log_a e$
12) $y = \ln(x)$	$y' = \frac{1}{x}$
13) $y = a^x$	$y' = a^x \ln(a)$
14) $y = e^x$	$y' = e^x$
15) $y = \text{sen}(x)$	$y' = cos(x)$
16) $y = cos(x)$	$y' = -\text{sen}(x)$
17) $y = tg(x)$	$y' = \frac{1}{\cos^2(x)} = 1 + \text{tg}^2(x) = \sec^2(x)$
18) $y = cotg(x)$	$y' = \frac{-1}{\text{sen}^2(x)} = -\text{cosec}^2(x) = -(1 + \text{cotg}^2(x))$
19) $y = sec(x)$	$y' = sec(x).tg(x)$
20) $y = cosec(x)$	$y' = -cosec(x).cotg(x)$
21) $y = arcsen(x)$	$y' = \frac{1}{\sqrt{1-x^2}}$
22) $y = arccos(x)$	$y' = \frac{-1}{\sqrt{1-x^2}}$
23) $y = arctg(x)$	$y' = \frac{1}{1+x^2}$
24) $y = arccotg(x)$	$y' = \frac{-1}{1+x^2}$
25) $y = arcsec(x)$	$y' = \frac{1}{x\sqrt{x^2-1}}$
26) $y = arccosec(x)$	$y' = \frac{-1}{x\sqrt{x^2-1}}$

8) $y = u^m$	$y' = m.u^{m-1}.u'$
9) $y = \sqrt[m]{u}$	$y' = \frac{u'}{m\sqrt[m]{u^{m-1}}}$
10) $y = \sqrt{u}$	$y' = \frac{u'}{2\sqrt{u}}$
11) $y = \log_a(u)$	$y' = \frac{u'}{u}\log_a e$
12) $y = \ln(u)$	$y' = \frac{u'}{u}$
13) $y = a^u$	$y' = a^u.u'.\ln(a)$
14) $y = e^u$	$y' = u'.e^u$
15) $y = \text{sen}(u)$	$y' = cos(u).u'$
16) $y = \cos(u)$	$y' = -\text{sen}(u).u'$
17) $y = \text{tg}(u)$	$y' = \frac{u'}{\cos^2(u)} = u'(1 + \text{tg}^2(u)) = \sec^2(u).u'$
18) $y = \text{cotg}(u)$	$y' = \frac{-u'}{\text{sen}^2(u)} = -u'(1 + \text{cotg}^2(u)) = -\text{cosec}^2(u).u'$
19) $y = \text{sec}(u)$	$y' = sec(u).tg(u).u'$
20) $y = \text{cosec}(u)$	$y' = -cosec(u).cotg(u).u'$
21) $y = \text{arcsen}(u)$	$y' = \frac{u'}{\sqrt{1-u^2}}$
22) $y = \arccos(u)$	$y' = \frac{-u'}{\sqrt{1-u^2}}$
23) $y = \text{arctg}(u)$	$y' = \frac{u'}{1+u^2}$
24) $y = \text{arccotg}(u)$	$y' = \frac{-u'}{1+u^2}$
25) $y = \text{arcsec}(u)$	$y' = \frac{u'}{u\sqrt{u^2-1}}$
26) $y = \text{arccosec}(u)$	$y' = \frac{-u'}{u\sqrt{u^2-1}}$

TABLA DE DERIVADAS E INTEGRALES

INTEGRALES

Integrales Inmediatas de una Variable

1) $\int (f_1(x) \pm f_2(x) \pm \cdots \pm f_n(x))dx = \int f_1(x)dx \pm \int f_2(x) \pm \cdots \pm \int f_n(x)dx$

3) $\int dx = x + c$	

Integrales Inmediatas de Funciones

$u = f(x)$

2) $\int u \,.\, dv = u.v - \int v.du$ *Integración por partes*

4) $\int x^m.dx = \frac{x^{m+1}}{m+1} + c$	4) $\int u^m.u'.dx = \frac{u^{m+1}}{m+1} + c$
5) $\int \frac{1}{x}.dx = \ln\|x\| + c$	5) $\int \frac{u'}{u}.dx = \ln\|u\| + c$
6) $\int e^x.dx = e^x + c$	6) $\int e^u.u'.dx = e^u + c$
7) $\int a^x.dx = \frac{a^x}{\ln(a)} + c$	7) $\int a^u.u'.dx = \frac{a^u}{\ln(a)} + c$
8) $\int \text{sen}(x).dx = -cos(x) + c$	8) $\int \text{sen}(u).u'.dx = -cos(u) + c$
9) $\int \cos(x).dx = \text{sen}(x) + c$	9) $\int \cos(u).u'.dx = \text{sen}(u) + c$
10) $\int \text{tg}(x).dx = \ln\|\sec(x)\| + c = -\ln\|\cos(x)\| + c$	10) $\int \text{tg}(u).u'.dx = \ln\|\sec(u)\| + c = -\ln\|\cos(u)\| + c$
11) $\int \text{cosec}(x).dx = \ln\|\text{cosec}(x) - cotg(x)\| + c$	11) $\int \text{cosec}(x).u'.dx = \ln\|\text{cosec}(u) - cotg(u)\| + c$
12) $\int \sec(x).dx = \ln\|\sec(x) + tg(x)\| + c$	12) $\int \sec(u).u'.dx = \ln\|\sec(u) + tg(u)\| + c$
13) $\int \text{cotg}(x).dx = \ln\|sen(x)\| + c$	13) $\int \text{cotg}(u).u'.dx = \ln\|sen(u)\| + c$
14) $\int \frac{1}{\cos^2(x)}.dx = \int \sec^2(x).dx = tg(x) + c$	14) $\int \frac{u'}{\cos^2(u)}.dx = \int \sec^2(u).dx = tg(u) + c$
15) $\int \frac{1}{\text{sen}^2(x)}.dx = \int \text{cosec}^2(x).dx = -cotg(x) + c$	15) $\int \frac{u'}{\text{sen}^2(u)}.dx = \int \text{cosec}^2(u).dx = -cotg(u) + c$
16) $\int \sec(x).tg(x)dx = \sec(x) + c$	16) $\int \sec(u).tg(u).u'.dx = \sec(u) + c$
17) $\int \text{cosec}(x).cotg(x)dx = -\text{cosec}(x) + c$	17) $\int \text{cosec}(u).cotg(u).u'.dx = -\text{cosec}(u) + c$
18) $\int \frac{1}{\sqrt{1-x^2}}.dx = \begin{cases} arcsen(x) + c \\ -\arccos(x) + c \end{cases}$	18) $\int \frac{1}{\sqrt{1-u^2}}.u'.dx = \begin{cases} arcsen(u) + c \\ -\arccos(u) + c \end{cases}$
19) $\int \frac{1}{1+x^2}.dx = \begin{cases} arctg(x) + c \\ -\text{arccotg}(x) + c \end{cases}$	19) $\int \frac{1}{1+u^2}.u'.dx = \begin{cases} arctg(u) + c \\ -\text{arccotg}(u) + c \end{cases}$
20) $\int \frac{1}{x\sqrt{x^2-1}}.dx = \begin{cases} arcsec(x) + c \\ -\text{arccosec}(x) + c \end{cases}$	20) $\int \frac{1}{u\sqrt{u^2-1}}.u'.dx = \begin{cases} arcsec(u) + c \\ -\text{arccosec}(u) + c \end{cases}$

RELACIONES TRIGONOMÉTRICAS

Suma y Diferencia de Ángulos

$sen(\alpha + \beta) = sen\alpha \,.\, cos\beta + cos\alpha \,.\, sen\beta$

$sen(\alpha - \beta) = sen\alpha \,.\, cos\beta - cos\alpha \,.\, sen\beta$

$cos(\alpha + \beta) = cos\alpha \,.\, cos\beta - sen\alpha \,.\, sen\beta$

$cos(\alpha + \beta) = cos\alpha \,.\, cos\beta + sen\alpha \,.\, sen\beta$

$tg\,(\alpha + \beta) = \frac{tg\alpha + tg\beta}{1 - tg\alpha \,.\, tg\beta}$

$tg\,(\alpha - \beta) = \frac{tg\alpha - tg\beta}{1 + tg\alpha \,.\, tg\beta}$

Identidades de Ángulos

$sen(2\alpha) = 2.sen\alpha.cos\alpha$

$\cos(2\alpha) = cos^2\alpha - sen^2\alpha$

$\text{tg}(2\alpha) = \frac{2.tg\ \alpha}{1 - tg^2\alpha}$

Identidad Pitagórica

$sen^2\alpha + cos^2\alpha = 1$

Suma a Producto

$$\text{sen}\,\alpha \pm \text{sen}\,\beta = 2\,\text{sen}\frac{1}{2}(\alpha \pm \beta)\cos\frac{1}{2}(\alpha \mp \beta)$$

$$\cos\alpha + \cos\beta = 2\cos\frac{1}{2}(\alpha + \beta)\cos\frac{1}{2}(\alpha - \beta)$$

$$\cos\alpha - \cos\beta = -2\,\text{sen}\frac{1}{2}(\alpha + \beta)\cos\frac{1}{2}(\alpha - \beta)$$